

LEXAN™ FR Resin 923A - Asia

Polycarbonate
SABIC

PROSPECTOR®

www.ulprospector.com

Technical Data

Product Description

LEXAN™ 923A resin is a 15 MFR polycarbonate. UV stabilized. Mold release. Flame retardant, UL94 V0 rated. UL746C f1 rated. Available in transparent and translucent colors.

General

Material Status	• Commercial: Active		
UL Yellow Card ¹	• E207780-100726468		
Search for UL Yellow Card	• SABIC		
Availability	• Asia Pacific		
Uses	• Aerospace Applications • Appliances • Automotive Exterior Parts • Construction Applications • Electric Vehicle Applications	• Electrical Parts • Electrical/Electronic Applications • Electronic Displays • Lawn and Garden Equipment • Lighting Applications	• Medical/Healthcare Applications • Military/Defense Applications • Outdoor Applications • Personal Care • Water Management
Also Available In	• Europe	• Latin America	• North America

Physical	Nominal Value Unit	Test Method
Density / Specific Gravity	1.22 1.21 g/cm³	ASTM D792
Specific Volume	0.830 cm³/g	ASTM D792
Melt Mass-Flow Rate (MFR) (300°C/1.2 kg)	15 g/10 min	ASTM D1238
Molding Shrinkage - Flow (3.20 mm)	0.50 to 0.70 %	Internal Method
Water Absorption		ASTM D570
24 hr, 23°C	0.15 %	
Saturation, 23°C	0.35 %	
Equilibrium, 100°C	0.58 %	
Outdoor Suitability	f1	UL 746C
Mechanical	Nominal Value Unit	Test Method
Tensile Strength ³		ASTM D638
Yield	62.0 MPa	
Break	55.0 MPa	
Tensile Elongation ³		ASTM D638
Yield	7.0 %	
Break	90 %	
Flexural Modulus ⁴ (50.0 mm Span)	2240 MPa	ASTM D790
Flexural Strength ⁴ (Yield, 50.0 mm Span)	91.0 MPa	ASTM D790
Taber Abrasion Resistance		ASTM D1044
1000 Cycles, 1000 g, CS-17 Wheel	10.0 mg	
Impact	Nominal Value Unit	Test Method
Notched Izod Impact (23°C)	640 J/m	ASTM D256
Unnotched Izod Impact (23°C)	3200 J/m	ASTM D4812
Gardner Impact (23°C)	169 J	ASTM D3029
Tensile Impact Strength ⁵	525 kJ/m²	ASTM D1822
Hardness	Nominal Value Unit	Test Method
Rockwell Hardness		ASTM D785
M-Scale	70	
R-Scale	118	



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Thermal	Nominal Value Unit	Test Method
Deflection Temperature Under Load		ASTM D648
0.45 MPa, Unannealed, 6.40 mm	137 °C	
1.8 MPa, Unannealed, 6.40 mm	132 °C	
Vicat Softening Temperature	151 °C	ASTM D1525 ⁶
CLTE - Flow (-40 to 95°C)	6.8E-5 cm/cm/°C	ASTM E831
Thermal Conductivity	0.19 W/m/K	ASTM C177
RTI Elec	130 °C	UL 746B
RTI Imp	120 °C	UL 746B
RTI Str	130 °C	UL 746B
Electrical	Nominal Value Unit	Test Method
Volume Resistivity	> 1.0E+17 ohms·cm	ASTM D257
Dielectric Strength (3.20 mm, in Air)	17 kV/mm	ASTM D149
Dielectric Constant		ASTM D150
60 Hz	3.01	
50 kHz	3.01	
1 MHz	2.96	
Dissipation Factor		ASTM D150
50 Hz	9.0E-4	
60 Hz	9.0E-4	
1 MHz	0.010	
Arc Resistance ⁷	PLC 7	ASTM D495
Comparative Tracking Index (CTI)	PLC 3	UL 746A
High Amp Arc Ignition (HAI) ⁸	PLC 2	UL 746A
High Voltage Arc Resistance to Ignition (HVAR)	PLC 3	UL 746A
Hot-wire Ignition (HWI)	PLC 2	UL 746A
Flammability	Nominal Value Unit	Test Method
Flame Rating		UL 94
1.5 mm	V-2	
3.0 mm	V-0	
Optical	Nominal Value Unit	Test Method
Refractive Index	1.586	ASTM D542
Light Transmittance (2540 µm)	85.0 %	ASTM D1003
Haze (2540 µm)	1.00 %	ASTM D1003
Injection	Nominal Value Unit	
Drying Temperature	120 °C	
Drying Time	3.0 to 4.0 hr	
Suggested Max Moisture	0.020 %	
Suggested Shot Size	40 to 60 %	
Rear Temperature	265 to 290 °C	
Middle Temperature	275 to 300 °C	
Front Temperature	290 to 310 °C	
Nozzle Temperature	280 to 305 °C	
Processing (Melt) Temp	290 to 310 °C	
Mold Temperature	70 to 95 °C	
Back Pressure	0.300 to 0.700 MPa	
Screw Speed	40 to 70 rpm	
Vent Depth	0.025 to 0.076 mm	

Injection Notes

- Drying Time (Cumulative): 48 hr



Notes

¹ A UL Yellow Card contains UL-verified flammability and electrical characteristics. UL Prospector continually works to link Yellow Cards to individual plastic materials in Prospector, however this list may not include all of the appropriate links. It is important that you verify the association between these Yellow Cards and the plastic material found in Prospector. For a complete listing of Yellow Cards, visit the UL Yellow Card Search.

² Typical properties: these are not to be construed as specifications.

³ Type I, 50 mm/min

⁴ 1.3 mm/min

⁵ Type S

⁶ Rate A (50°C/h), Loading 2 (50 N)

⁷ Tungsten Electrode

⁸ Surface



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Where to Buy

Supplier

SABIC

Web: <http://www.sabic.com/>

Distributor

3Polymer (Guangzhou) Chemical Technology Co., Ltd.

Telephone: +86-20-3466-7988

Web: <http://3polymer.com>

Availability: China

